

EYUM

Eye-Gaze Communication App

QUICK START GUIDE · 5-Minute Essentials

Once the app is installed on a smartphone or tablet, the user launches it and is shown a disclaimer and a short set of instructions. After accepting the disclaimer, the user begins a session.

The camera preview should show the eye region of the face clearly — this confirms the app is tracking correctly. From there, the user selects a phrase from the phrase board using eye movements alone.

Every phrase on the board is mapped to a binary codeword. The user builds that codeword one digit at a time. Do not look at the left or right side of the screen — turn the eyeballs themselves as far as they will go, as if glancing sideways at something past the edge of the device. Turning them fully **LEFT** enters **0**, turning them fully **RIGHT** enters **1**. Holding the gaze at the **CENTER** for two seconds ends the entry, and the device speaks the selected phrase aloud. When the same digit repeats — 00, for example — move the eyes to that side, come briefly back to the centre, then move to the same side again.

No special hardware. No keyboard. No spelling. Just a Galaxy phone or tablet and its front camera.

The three gaze movements

Look LEFT	= 0	Turn the eyes fully left, then back. Adds the digit 0
Look RIGHT	= 1	Turn the eyes fully right, then back. Adds the digit 1
Look CENTER	2 sec	Confirms — the phrase is spoken aloud

Example: "Uncomfortable" = 011 → LEFT, RIGHT, centre (briefly), RIGHT, then CENTER for 2 seconds. The short return to the centre separates the two RIGHT turns; consecutive different directions need no pause.



LEFT (0)



RIGHT (1)



RIGHT (1)



CENTER (2 s)

Made a mistake? Just look at the center for 2 seconds. Codes that do not exist are thrown away automatically, and you can start again.

Safety: This is a communication aid, not a medical device. A caregiver must always be present, and there must always be a backup way to call for help.

1. Setting Up



Mounting position

Position the device

- About 30 cm from the user's face
- Camera at eye level, slightly above the line of sight so it does not block the view
- Face centred in the camera preview
- Light in FRONT of the user — never a window behind them
- Wipe the camera lens before use

Glasses: remove them if possible. If they must be worn, tilt the device until no glare shows on the lenses in the camera preview — reflections block eye detection.

First launch — 3 taps

1	Choose the language at the top of the screen: English or 한국어. This sets the menus, the phrase list and the voice.
2	Read the disclaimer and tap Accept.
3	On the Instructions screen choose the device type — Galaxy Phone or Galaxy Tab — so the app knows where the camera is.

The three buttons

Button	What it does
Start	Turns the camera on and starts tracking. The status light turns GREEN (TRACKING).
Pause / Resume	Suspends tracking; the camera stays on. Status light turns ORANGE (PAUSED).
Stop	Turns the camera off and stops everything.

The user can pause without help. Closing the eyes for five seconds or longer pauses tracking; closing them again for five seconds resumes it. Use this while resting or sleeping, or whenever the user is not composing a phrase so that ordinary eye movements are not read as input

Which orientation?

- Galaxy Tab: use LANDSCAPE. In portrait the camera sits far to the side and accuracy drops.
- Galaxy Phone: use PORTRAIT. The camera sits at the top centre of the screen, so the left and right gaze directions are detected evenly.

Before the user starts: the caregiver should try the app first. Ten minutes of practice makes teaching much easier.

2. The Main Phrase Chart

This is the only page most people need. Short codes are the most urgent phrases — emergencies take a single look.

Code	Gaze	Says
0	L → centre	Can't breathe
1	R → centre	Suction
00	L L → centre	Yes
01	L R → centre	No
10	R L → centre	Repeat
11	R R → centre	Wait
000	L L L → centre	Thirsty ★
001	L L R → centre	Urinate ★
010	L R L → centre	Defecate ★
011	L R R → centre	Uncomfortable ★
100	R L L → centre	opens the PAIN page
101	R L R → centre	opens the ACTION page
110	R R L → centre	opens the ENVIRON · STATE page
111	R R R → centre	opens the FEELING page

Red	Emergency — one look only
Green	Yes / No / Repeat / Wait — answering questions
Yellow	★ Shortcuts — these four slots change by themselves (see below)
Blue	Doorways to the four category pages

★ The yellow shortcuts learn from use

- Every time a phrase is chosen from a category page, its counter goes up by one.
- The four most-used phrases move into the yellow slots (000–011), so frequent needs get shorter codes. The board never changes during a session: the slots are recalculated only when Stop is pressed or the app is restarted.
- Until enough phrases have been used, the defaults stay: Thirsty, Urinate, Defecate, Uncomfortable.
- To see or clear the counters: Setup → Zone 3A – Usage Frequency → Show / Reset all usage counts.

Teaching order: practise the red and green phrases first (1–2 looks) until they are reliable. Add the yellow shortcuts next, then introduce the category pages one at a time. Sessions of 10–15 minutes work better than long ones.

3. The Four Category Pages

Entering a 3-digit code opens a category page. The counter then resets to empty, and the user enters a second, shorter code to select a phrase within that category. After the phrase is spoken the app returns to Main by itself. Looking LEFT + centre (code 0) always goes back to Main.

PAIN enter code 100 (R L L)		ACTION enter code 101 (R L R)	
0	← back to Main	0	← back to Main
1	Stomachache	1	Medicine
00	Headache	00	Change posture
01	Back pain	01	Turn me
10	Neck pain	10	Lift head
11	Chest pain	11	Lower head
000	Nausea	000	Wipe
001	Mouth dry	001	Cover me
010	Eyes dry	010	Call nurse
011	Leg pain	011	Come here
100	Shoulder pain	100	Sit me down
101	Cramp	101	Hold me still
110	Numb	110	Urinate
111	Getting worse	111	Defecate

ENVIRON · STATE enter code 110 (R R L)		FEELING enter code 111 (R R R)	
0	← back to Main	0	← back to Main
1	Cold	1	Thank you
00	Hot	00	Sorry
01	Too bright	01	Anxious
10	Too dark	10	Scared
11	Too loud	11	Sad
000	Sleepy	000	Lonely
001	Itchy	001	Happy
010	Thirsty	010	Exhausted
011	Uncomfortable	011	Angry
100	Open window	100	Miss you
101	Close window	101	I love you
110	Turn on TV	110	Proud of you
111	Turn off TV	111	Don't worry

Adding your own phrases: Instructions screen → Setup → tap any slot → type the phrase → Save, then Save & Return. Keep phrases to 1–3 words so the voice stays clear. To undo all edits at once, use Restore default phrases at the bottom of the Setup screen. Three items cannot be changed: "Can't breathe" (code 0), "back to Main" (code 0 on category pages), and the four category doorways (100–111).

4. Four Worked Examples

To say	Do this
"Can't breathe"	LEFT → centre 2 s. Spoken immediately.
"Headache"	RIGHT LEFT LEFT → centre 2 s (PAIN page opens) → LEFT LEFT → centre 2 s.
"Thank you"	RIGHT RIGHT RIGHT → centre 2 s (FEELING page opens) → RIGHT → centre 2 s.
Go back to Main	LEFT → centre 2 s. Nothing is spoken.

5. If Something Is Not Working

Problem	What to do
Camera will not start	Android Settings → Apps → EYEUM → Permissions → allow Camera.
No voice	Check the volume buttons. Then Android Settings → General Management → Language & Input → Text-to-Speech.
Gaze not detected	More light. Clean the lens. Move the device closer. Remove glasses, or tilt the device until the glare disappears.
Reacts to every small movement	Check the eyelids are fully open, not half-closed. Ordinary small eye movements should not enter a digit.
Wrong phrase was spoken	Use Repeat (10) or Wait (11) to signal the caregiver, then enter the intended code again.
Unexpected phrases in the yellow slots	Setup → Zone 3A – Usage Frequency → Reset all usage counts.
Phrases in the wrong language	Go back to the Disclaimer screen and select the language again.

Safety notice

This application is a communication aid only. It is not a medical device and must never be the only way the user can alert someone in an emergency.

- Use only with a caregiver present and monitoring.
- Always keep a backup alert method within reach.
- Have a speech-language pathologist assess the user before long-term use.

The developers accept no liability for consequences arising from use of this application. By using it you accept these risks.

This is a short guide. The full User Manual contains the complete Setup screen reference, screen layouts for every device and orientation, and detailed training guidance.

Questions, problems, or feedback: contact.eyum@gmail.com